NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

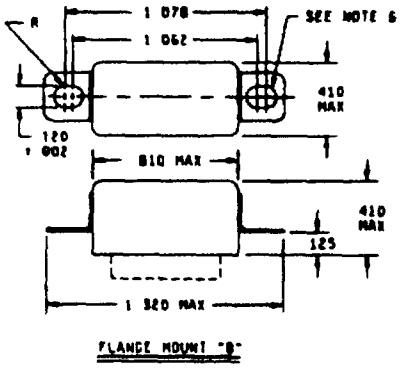
MIL-PRF-39016/6K
15 February 1991
SUPERSEDING
MIL-R-39016/6J
26 September 1988

PERFORMANCE SPECIFICATION SHEET

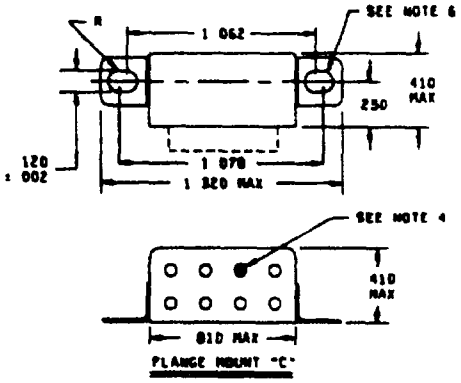
RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT,
LOW LEVEL TO 2 AMPERES (0.200-INCH TERMINAL SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.

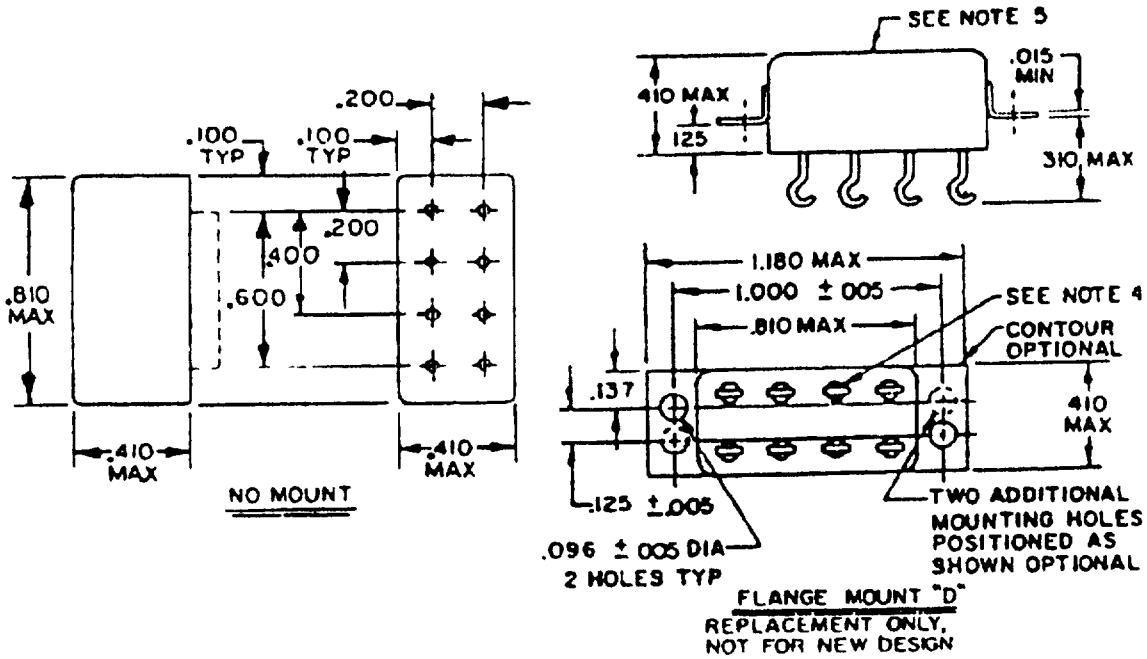


FLANGE MOUNT "B"

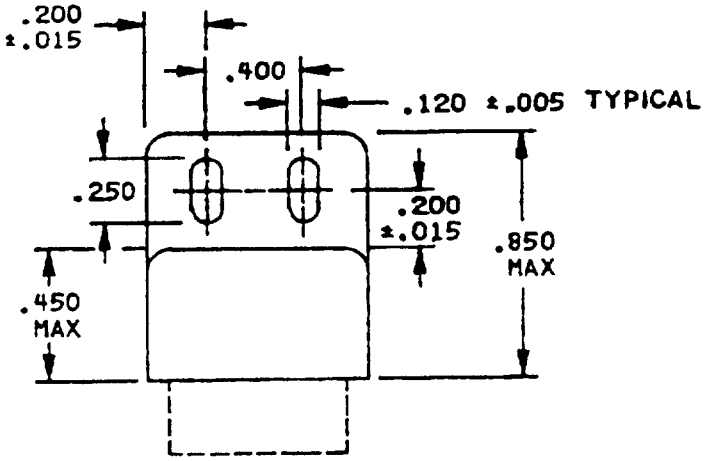


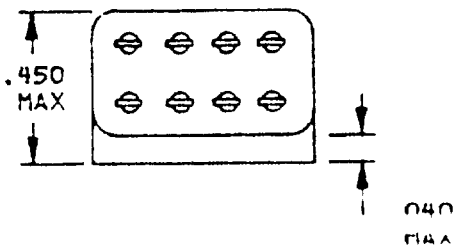
FLANGE MOUNT "C"

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FLANGE MOUNT "D"
REPLACEMENT ONLY, NOT FOR NEW DESIGN

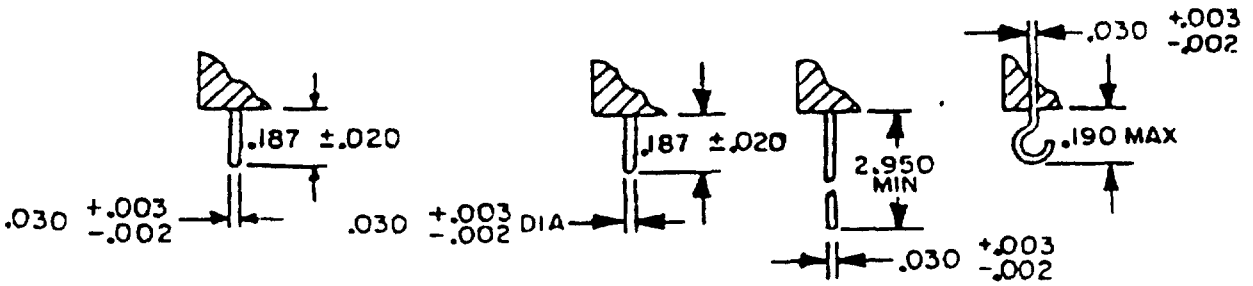


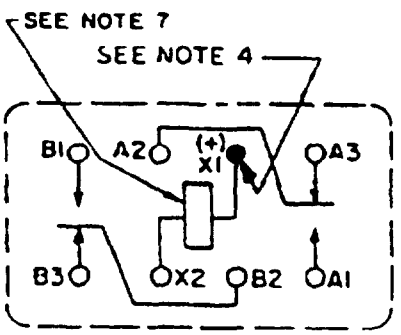


FLANGE MOUNT "E"

FIGURE 1. Dimensions and configuration - Cont'd

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Inches	mm	Inches	mm
.002	0.05	.250	6.35
.003	0.08	.310	7.87
.005	0.13	.400	10.10
.015	0.38	.410	10.41
.020	0.51	.450	11.43
.030	0.76	.600	15.24
.040	1.02	.810	20.57
.096	2.44	.850	21.59
100	2 54	1 000	25 40
120	3.05	1 062	26.97
.125	3.18	1.078	27.38
.137	3.48	1.180	29.97
.187	4.75	1 320	33.53
.190	4.83	2.950	74.93
.200	5.08		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Indicated terminal shall be marked with a contrasting bead.
5. Circuit diagram (including terminal designations) shall be marked on top if legible from the mounted position, otherwise marking surface is optional.
6. Mounting screw head clearances are based on use of .112 diameter fillister head screws.
7. Coil symbol shall be in accordance with MIL-STD-1285
8. Mounting surface finish shall be compatible with aluminum (duralumin type) as defined by the compatible couples table of MIL-R-39016.
9. Finish. Gold plate in accordance with MIL-G-45204, type II, class 1, grade C, except knoop hardness shall be 130 to 240. Underplating: Nickel, 50 to 150 microinches thick

FIGURE 1. Dimensions and configuration - Continued

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REQUIREMENTS:

CONTACT DATA:

Load ratings.	Type 1 1/	Type 2 1/
High level (relay case grounded):		
Resistive:	2 A at 28 V dc 0.3 A at 115 V ac (60 and 400 Hz) 2/	2 A at 28 V dc 0.1 A at 115 V ac (60 and 400 Hz) 2/
Inductive.	0.75 A at 28 V dc with 0.200 H inductance	0.5 A at 28 V dc with 0.200 H inductance
Lamp:	0.160 A at 28 V dc	0.160 A at 28 V dc
Low level:	10 to 50 µA at 10 to 50 mV dc or peak ac	10 to 50 µA at 10 to 50 mV dc or peak ac
Intermediate current:	Applicable	Applicable

All relays

Contact resistance and voltage drop

High level:

Before life: 0.050Ω maximum.
During life: Not more than 5 percent of open circuit voltage.
After life: 0.100Ω maximum.

Low level:

Before life: 0.050Ω maximum.
During life: 3Ω maximum.
After life: 0.150Ω maximum.

Intermediate current

Before. 0.050Ω maximum.
During: 1Ω maximum
After: 0.100Ω maximum.

Contact bounce: 2.0 ms maximum (applicable to failure rate level L).

Contact stabilization time: 2.0 ms maximum (applicable to failure rate levels M, P, and R)

Overload (high level only) Two times rated current

COIL DATA: See table I

Operate time 4 ms maximum over temperature range with rated coil voltage

Release time 4 ms maximum over temperature range from rated coil voltage

ELECTRICAL DATA

Insulation resistance 10,000 MΩ minimum

1/ Original equipment manufacturers (OEM's) should use type 2 relays whenever the ac resistance or dc inductive load ratings are satisfactory
2/ 400 Hz load testing is required for qualification only

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TABLE I. Dash numbers and applicable characteristics. 1/

Dash numbers 2/						Coil voltage 4/ (V dc)		At +25°C				Over temperature range 3/		
Wire lead (PI) 6/	Solder lug	Wire lead (SP)	Wire lead	Mount	Load rating	Rated	Max	Coil resistance ohms ±10%	Specified pick-up voltage (V dc) (max) 3/	Specified hold voltage (V dc) 3/	Specified drop-out voltage (V dc) 3/	Specified pick-up voltage (V dc) max	Specified hold voltage (V dc)	Specified drop-out voltage (V dc)
141	104	105	106	B										
---	107	125	108	C										
---	128	---	---	D 5/	Type 1									
142	129	109	110	None										
	149			E		26.5	32.0	700	13.5	8.0	1.5	18.0	14.0	1.0
241	204	205	206	B										
---	207	225	208	C										
---	228	---	---	D 5/	Type 2									
242	229	209	210	None										
	249			E										
143	111	112	113	B										
---	114	126	115	C	Type 1									
144	130	116	117	None										
	150			E		12.0	15.0	160	6.4	4.0	0 7	9 0	5.8	0 50
243	211	212	213	B										
---	214	226	215	C	Type 2									
244	230	216	217	None										
	250			E										
145	118	119	120	B										
---	121	127	122	C	Type 1									
146	131	123	124	None										
	151			E		6.0	7.5	40	3.2	2.0	0 35	4 5	2.9	0.25
245	218	219	220	B										
---	221	227	222	C	Type 2									
246	231	223	224	None										
	251			E										
147	132	133	134	B										
---	135	136	137	C	Type 1									
148	138	139	140	None										
	152			E		5 0	6 0	27	2 7	1 65	0 29	3 8	2 4	0 21
247	232	233	234	B										
---	235	236	237	C	Type 2									
248	238	239	240	None										
	252			E										

See footnotes on next page

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TABLE I. Dash numbers and applicable characteristics - Continued 1/

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum, or peak ac open circuit, are not recommended for subsequent use in low level applications.
2/ The suffix letter L, M, P, or R to designate the failure rate level shall be added to the applicable dash number. Failure rate level (percent per 10,000 operations). L, 3.0, M, 1.0; P, 0.1, R, 0.01. Example, 204M- - - -204R.
3/ Pick-up, hold and dropout voltages as shown are for test purposes only and are not to be used as design criteria.
4/ Caution: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.
5/ Not for new design.
6/ Solderability for plug-in relays is not applicable.

Dielectric withstanding voltage:

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized positions	1,000	
Between case, frame, or enclosure and coil	500	
Between all contacts and coil	1,000	350
Between open contacts in the energized and deenergized positions	500	All terminals to case
Between contact poles	1,000	

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 μs maximum for closed contacts, and 1 μs maximum closure for open contacts.

Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 μs maximum for closed contacts, and 1 μs maximum closure for open contacts (Applicable to qualification and group C testing only).

Shock (specified pulse) MIL-STD-202, method 213, test condition C (100 G). Contact chatter shall not exceed 10 μs maximum for closed contacts, and 1 μs maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat Applicable.

Acceleration: Applicable

PHYSICAL DATA:

Terminals See figure 1 and table I

Terminal strength 3 to 3 pounds pull

Terminal solderbility Applicable

Terminal twist test Applicable to wire lead terminals

Dimensions and configuration See figure 1

Weight 13 grams (0.46 ounce) maximum

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Identification marking (full): Applicable.

Seal: Hermetic.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles.

Low level. 100,000 cycles plus 900,000 cycles mechanical life.

(K) Part or Identifying Number (PIN): M39016/6- (plus dash number from table I and suffix letter designating failure rate level).

(K) QUALIFICATION INSPECTION AND SAMPLE SIZE: See table II.

TABLE II. Qualification inspection and sample size.

Single submission	Group submission		
	Type 1	Type 2	
33 units plus 1 open unit for level M at C = 0 1/ Qualification inspection as applicable.	M39016/6-104	M39016/6-204	33 units plus 1 open unit for level M at C = 0 1/ Qualification inspection as applicable.
	M3916/6-115 M39016/6-149	M39016/6-215 M39016/6-249	2 units, qualification inspection table, group II, and shock, vibration, acceleration, terminal strength, and seal.
	M39016/6-141 M39016/6-142	M39016/6-241 M39016/6-242	2 units, qualification inspection table, group II, shock, and vibration.
	M39016/6-123 M39016/6-132	M39016/6-223 M39016/6-236	2 units, qualification inspection table, group II
	For extension of qualification to type 2	For extension of qualification to type 1	
	M39016/6-204	Not applicable	1 unsealed sample unit

1/ The number of units required for qualification testing shall be increased as required in group V, table II, MIL-R-39016, if the contractor elects to test the number of units permitting one or more failures.

(K) QUALIFICATION INSPECTION (REDUCED TESTING) AND SAMPLE SIZE: See table III. If the relays produced for MIL-R-39016/6 are similar in construction and design to the relays produced for MIL-R-39016/55 (except for the diodes and coils, as applicable), then reduced testing for qualification of MIL-R-39016/6 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/55.

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Ⓚ TABLE III. Qualification inspection (reduced testing) and sample size

Examination or test
Group II of qualification inspection table (2 units each coil voltage)
1 unsealed sample unit for internal examination (submitted with test report)

Ⓚ SUPERSESSION DATA: See table IV.

Ⓚ TABLE IV. Supersession data. 1/

Dash number (type 1) PIN M39016/6-	Dash number (type 2) PIN M39016/6-	Superseded PIN MIL-R-5757/9	Superseded PIN MIL-R-5757/10-
-104	-204	-005, -006	005, 011, 017
-105	-205	-007, -008, -057, 058	006, 012, 018
-106	-206	-009, -010	---
-107	-207	-011, -012	001, 007, 013
-108	-208	-015, -016	002, 008, 014
-109	-209	-003, -004, -061, -062	062
-110	-210	-019, -020	---
-111	-211	-021, -022	---
-112	-212	-023, -024, -063, -064	---
-113	-213	-025, -026	---
-114	-214	-027, -028	---
-115	-215	-031, -032	---
-116	-216	-035, -036, -067, -068	---
-117	-217	-037, -038	---
-118	-218	-039, -040	---
-119	-219	-041, -042, -069, -070	---
-120	-220	-043, -044	---
-121	-221	-045, -046	---
-122	-222	-049, -050	---
-123	-223	-053, -054, -073, -074	-61
-124	-224	-055, -056	---
-125	-225	-013, -014, -059, -060	---
-126	-226	-029, -030, -065, -066	---
-127	-227	-047, -048, -071, -072	---
-128	-228	-001, -002	-019, 021
-129	-229	-017, -018	---
-130	-230	-033, -034	---
-131	-231	-051, -052	---
-132	-232	---	---
-133	-233	---	---
-134	-234	---	---
-135	-235	---	---
-136	-236	---	---
-137	-237	---	---
-138	-238	---	---
-139	-239	---	---
-140	-240	---	---
-141	-241	---	---
-142	-242	---	---
-143	-243	---	---

See footnote at end of table

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Ⓚ TABLE IV. Supersession data - Continued. 1/

Dash number (type 1) PIN M39016/6-	Dash number (type 2) PIN M39016/6-	Superseded PIN MIL-R-5757/9	Superseded PIN MIL-R-5757/10-
-144	-244	---	---
-145	-245	---	---
-146	-246	---	---
-147	-247	---	---
-148	-248	---	---
-149	-249	---	---
-150	-250	---	---
-151	-251	---	---
-152	-252	---	---

1/ The listed M39016/6-100 series PIN's may be substituted for the M39016/6-200 series, the MS757/9 and the MS757/10 PIN's. However, the M39016/6-200 series, the MS757/9 and the MS757/10 PIN's shall not be substituted for the M39016-100 series PIN's.

Ⓚ CROSS REFERENCE FOR GOVERNMENT SUPPORT: See table V.

Ⓚ TABLE V. Cross reference for Government logistical support.

Superseded PIN MS757/9-	Superseded PIN MS757/10-	Superseded PIN M39016/6-	Support with PIN M39016/6-
005, 006	005, 011, 017	-204	-104
007, 008, 057, 058	006, 012, 018	-205	-105
009, 010	---	-206	-106
011, 012	001, 007, 013	-207	-107
015, 016	002, 008, 014	-208	-108
003, 004, 061, 062	062	-209	-109
019, 020	---	-210	-110
021, 022	---	-211	-111
023, 024, 063, 064	---	-212	-112
025, 026	---	-213	-113
027, 028	---	-214	-114
031, 032	---	-215	-115
035, 036, 067, 068	---	-216	-116
037, 038	---	-217	-117
039, 040	---	-218	-118
041, 042, 069, 070	---	-219	-119
043, 044	---	-220	-120
045, 046	---	-221	-121
049, 050	---	-222	-122
053, 054, 073, 074	061	-223	-123
055, 056	-	-224	-124
013, 014, 059, 060		-225	-125
029, 030, 065, 066	---	-226	-126
047, 048, 071, 072	---	-227	-127
001, 002	019, 021	-228	-128

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Ⓚ TABLE V. Cross refernce for Government logistical support - Continued.

Superseded PIN M5757/9-	Superseded PIN M5757/10-	Superseded PIN M39016/6	Support with PIN M39016/6-
017, 018	---	-229	-129
033, 034	---	-230	-130
051, 052	---	-231	-131
---	---	-232	-132
---	---	-233	-133
---	---	-234	-134
---	---	-235	-135
---	---	-236	-136
---	---	-237	-137
---	---	-238	-138
---	---	-239	-139
---	---	-240	-140
---	---	-241	-141
---	---	-242	-142
---	---	-243	-143
---	---	-244	-144
---	---	-245	-145
---	---	-246	-146
---	---	-247	-147
---	---	-248	-148
---	---	-249	-149
---	---	-250	-150
---	---	-251	-151
---	---	-252	-152

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AR
Navy - AS
Air Force - 85, 99
DLA - ES

User activities
Navy - MC, OS, SH
Air Force - 19

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5945-0850)

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